

WASP-12b

R-0368

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231102 · created 2026-07-18T15:40:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460249.8431 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.022
Transit depth (Rp/Rs)^2	8.9e-08 +/- 0.00013 [%]
Orbital Inclination (inc)	80.76 +/- 2.89
Airmass coefficient 1 (a1)	1.128 +/- 0.029
Airmass coefficient 2 (a2)	-0.0954 +/- 0.0095
Scatter in the residuals of the lightcurve fit is	2.58 %
Best Comparison Star	None
Optimal Aperture	14.14
Optimal Annulus	44.77
Transit Duration (day)	0.103 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.022 vs 0.1178 (deviation 5.35σ)
Duration fit vs published	0.103 d vs 0.125 d (deviation 2.0σ)
Mid-time O–C	-1571.6 min
Depth SNR	0.0
Residual scatter	2.58 %

Light curve

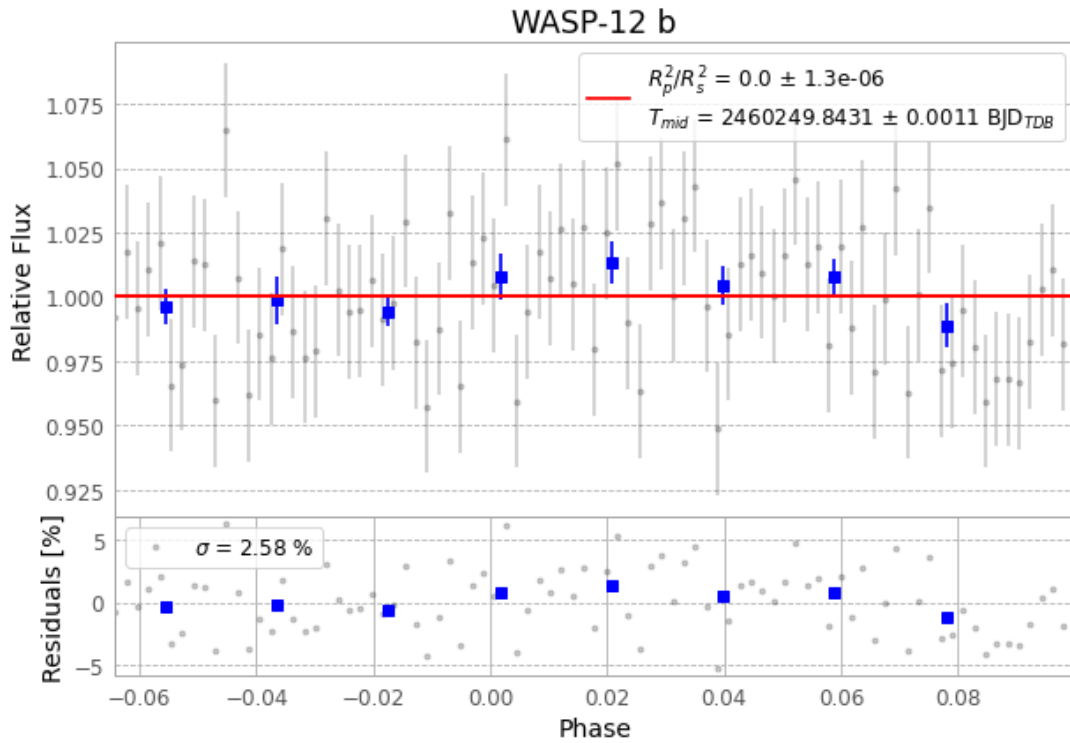


Figure 1 — FinalLightCurve_WASP-12 b_2023-10-31.png (SHA-256 b584d41c5334ebb0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 355], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1ac6d0e7d3a9b07c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

87 FITS frames (57 MB), WASP-12231101062715.FITS → WASP-12231101104515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231102.log (bd27d62183cc...), FinalLightCurve_WASP-12 b_2023-10-31.png (b584d41c5334...), FinalLightCurve_WASP-12 b_2023-10-31.csv (981f751e093a...)

Compute resources

Wall time 260.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 15:40Z.